

Chapter 11

High-dimensional Real Space and Mappings

*Mathematics consists in
proving the most obvious thing
in the least obvious way.*

George Polya, 1887-1985

In previous chapters, all of the functions that we studied are single-variable functions. However, the relationship between quantities should be described by using multi-variable functions in more general cases. Before studying the analytical properties of multi-variable functions, we need to be familiar with the structure of high-dimensional real space first.

In this chapter, we are devoted to study the algebraic structure, topological structure and Lebesgue measure on the set of high-dimensional real numbers.

11.1 Real Sets and Euclidian Space

Definition 11.1.1 Let $d \in \mathbb{N}_+$. The set of d -dimensional real numbers is defined by

$$\mathbb{R}^d = \{(x_1, x_2, \dots, x_d) : x_j \in \mathbb{R}, 1 \leq j \leq d\}.$$

For $P(x_1, x_2, \dots, x_d)$, $Q(y_1, y_2, \dots, y_d)$ in $\mathbb{R}^d$, the vector $\overrightarrow{PQ}$ is defined by

$$\overrightarrow{PQ} = (y_1 - x_1, y_2 - x_2, \dots, y_d - x_d).$$

We denote the origin of $\mathbb{R}^d$ by $O(0, 0, \dots, 0)$. And the elements in $\mathbb{R}^d$ can be either regarded as points $P(x_1, x_2, \dots, x_d)$ or vectors $\overrightarrow{OP} = (x_1, x_2, \dots, x_d)$ starting at O .

Then let us define two algebraic operations on $\mathbb{R}^d$.

Definition 11.1.2 $\forall \vec{x} = (x_1, x_2, \dots, x_d)$, $\vec{y} = (y_1, y_2, \dots, y_d) \in \mathbb{R}^d$ and $\forall c \in \mathbb{R}$, we define

1. $\vec{x} + \vec{y} = (x_1 + y_1, x_2 + y_2, \dots, x_d + y_d)$;
2. $c\vec{x} = (cx_1, cx_2, \dots, cx_d)$.

Equipped with the above two operations, $\mathbb{R}^d$ becomes a vector space on $\mathbb{R}$. This is the algebraic structure on $\mathbb{R}^d$. However, we can also give $\mathbb{R}^d$ geometric structure.

Definition 11.1.3 $\forall \vec{x} = (x_1, x_2, \dots, x_d), \vec{y} = (y_1, y_2, \dots, y_d) \in \mathbb{R}^d$, we define

$$\langle \vec{x}, \vec{y} \rangle = \sum_{k=1}^d x_k y_k.$$

It is very easy to check:

1. $\langle \vec{x}, \vec{x} \rangle \geq 0$, and $\langle \vec{x}, \vec{x} \rangle = 0$ if and only if $\vec{x} = \vec{0}$;
2. $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$;
3. $\forall c_1, c_2 \in \mathbb{R}, \langle c_1 \vec{x}_1 + c_2 \vec{x}_2, \vec{y} \rangle = c_1 \langle \vec{x}_1, \vec{y} \rangle + c_2 \langle \vec{x}_2, \vec{y} \rangle$.

In mathematics, a two-variable mapping $\langle \cdot, \cdot \rangle : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ satisfying the above three properties is called an inner product. And a vector space equipped with inner product is called Euclidian space. We will also use $\vec{x} \cdot \vec{y}$ to denote the inner product of $\vec{x}$ and $\vec{y}$ in this book.

Theorem 11.1.4 $\mathbb{R}^d$ is a Euclidian space.

In fact, inner product provide norm, distance and angle for vectors in $\mathbb{R}^d$.

Definition 11.1.5 We define the norm of $\vec{x} \in \mathbb{R}^d$ by

$$\|\vec{x}\| = \sqrt{\vec{x} \cdot \vec{x}} = \left(\sum_{k=1}^d x_k^2 \right)^{\frac{1}{2}}.$$

And the distance of $\vec{x}, \vec{y} \in \mathbb{R}^d$ is defined by

$$d(\vec{x}, \vec{y}) = \|\vec{x} - \vec{y}\| = \left(\sum_{k=1}^d (x_k - y_k)^2 \right)^{\frac{1}{2}}.$$

Given two points $P(x_1, x_2, \dots, x_d), Q(y_1, y_2, \dots, y_d)$ in $\mathbb{R}^d$, the distance between P and Q is then defined by

$$d(P, Q) = \|\overrightarrow{PQ}\| = \left(\sum_{k=1}^d (x_k - y_k)^2 \right)^{\frac{1}{2}}.$$

Suppose $\vec{x}$ and $\vec{y}$ are both nonzero, the angle between $\vec{x}$ and $\vec{y}$ is described by

$$\cos \langle \vec{x}, \vec{y} \rangle = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \cdot \|\vec{y}\|}$$

Remark 1. By Cauchy-Schwartz Inequality, $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \cdot \|\vec{y}\|$. Thus,

$$|\cos \langle \vec{x}, \vec{y} \rangle| \leq 1.$$

2. By Minkowski inequality, $d(P, Q) \leq d(P, R) + d(R, Q)$ for all $P, Q, R \in \mathbb{R}^d$.

In the end of this section, we want to introduce two operational laws of real sets which will be used frequently in the next several sections

Proposition 11.1.6 (Distributive Law) Let I be an index set, $E \subseteq \mathbb{R}^d$ and $\{S_\alpha : \alpha \in I\}$ be a family of subsets of $\mathbb{R}^d$. Then

$$E \cap \left(\bigcup_{\alpha \in I} S_\alpha \right) = \bigcup_{\alpha \in I} (E \cap S_\alpha) \quad \text{and} \quad E \cup \left(\bigcap_{\alpha \in I} S_\alpha \right) = \bigcap_{\alpha \in I} (E \cup S_\alpha).$$

Proof Since

$$\begin{aligned} \vec{x} \in E \cap \left(\bigcup_{\alpha \in I} S_\alpha \right) &\iff \vec{x} \in E \text{ and } \exists \alpha_0 \in I \text{ such that } \vec{x} \in S_{\alpha_0} \\ &\iff \vec{x} \in E \cap S_{\alpha_0} \text{ for some } \alpha_0 \in I \\ &\iff \vec{x} \in \bigcup_{\alpha \in I} (E \cap S_\alpha), \end{aligned}$$

we get $E \cap \left(\bigcup_{\alpha \in I} S_\alpha \right) = \bigcup_{\alpha \in I} (E \cap S_\alpha)$. By same way, $E \cup \left(\bigcap_{\alpha \in I} S_\alpha \right) = \bigcap_{\alpha \in I} (E \cup S_\alpha)$. $\square$

Proposition 11.1.7 (De Morgan's Law) Let I be an index set and $\{S_\alpha : \alpha \in I\}$ be a family of subsets of $\mathbb{R}^d$. Then

$$\left(\bigcup_{\alpha \in I} S_\alpha \right)^c = \bigcap_{\alpha \in I} S_\alpha^c \quad \text{and} \quad \left(\bigcap_{\alpha \in I} S_\alpha \right)^c = \bigcup_{\alpha \in I} S_\alpha^c.$$

Where $S^c = \mathbb{R}^d \setminus S$ is called the complementary set of S .

Proof Since

$$\begin{aligned} \vec{x} \in \left(\bigcup_{\alpha \in I} S_\alpha \right)^c &\iff \vec{x} \notin \bigcup_{\alpha \in I} S_\alpha \\ &\iff \forall \alpha \in I, \vec{x} \notin S_\alpha \\ &\iff \forall \alpha \in I, \vec{x} \in S_\alpha^c \\ &\iff \vec{x} \in \bigcap_{\alpha \in I} S_\alpha^c, \end{aligned}$$

we get $\left(\bigcup_{\alpha \in I} S_\alpha \right)^c = \bigcap_{\alpha \in I} S_\alpha^c$. By same way, $\left(\bigcap_{\alpha \in I} S_\alpha \right)^c = \bigcup_{\alpha \in I} S_\alpha^c$. $\square$

11.2 Topology on $\mathbb{R}^d$

Roughly speaking, topology on a set is rule defining which subsets are open sets. Since we have already defined distance on $\mathbb{R}^d$, the topology of $\mathbb{R}^d$ is naturally generated by its distance.

11.2.1 Structure of Open Sets

Definition 11.2.1 We say P is an interior point of $S \subseteq \mathbb{R}^d$, if $\exists r > 0$ such that

$$B_r(P) := \{Q \in \mathbb{R}^d : d(P, Q) < r\} \subseteq S.$$

We use $\overset{\circ}{S}$ to denote the set of interior points of S . And if $\overset{\circ}{S} = S$, then we say S is an open set in $\mathbb{R}^d$.

Example 11.2.2 (a, b) is open in $\mathbb{R} = \mathbb{R}^1$, $B_r(P)$ is open in $\mathbb{R}^d$. And $\mathbb{R}^d$ and $\emptyset$ are both open in $\mathbb{R}^d$.

Open sets in $\mathbb{R}^d$ also have the following properties.

Proposition 11.2.3 If A, B are both open in $\mathbb{R}^d$, then $A \cap B$ is open. And let I be an index set in $\mathbb{R}$. If $\{A_\alpha : \alpha \in I\}$ are all open in $\mathbb{R}^d$, then $\bigcup_{\alpha \in I} A_\alpha$ is also open.

Proof (1) Since A, B are both open in $\mathbb{R}^d$, $\forall P \in A \cap B$, $\exists r_1, r_2 > 0$ such that

$$B_{r_1}(P) \subseteq A \quad \text{and} \quad B_{r_2}(P) \subseteq B.$$

Set $r = \min\{r_1, r_2\}$, then $B_r(P) \subseteq A \cap B$. This means $A \cap B$ is open.

(2) $\forall P \in \bigcup_{\alpha \in I} A_\alpha$, $\exists \alpha_0 \in I$ such that $P \in A_{\alpha_0}$. Since A_{α_0} is open, $\exists r > 0$ such that

$$B_r(P) \subseteq A_{\alpha_0} \subseteq \bigcup_{\alpha \in I} A_\alpha.$$

This means P is an interior point of $\bigcup_{\alpha \in I} A_\alpha$. Thus, $\bigcup_{\alpha \in I} A_\alpha$ is open. $\square$

Remark By mathematical induction, we can also get the following consequence. If $A_1, A_2, \dots, A_n$ are finitely many open sets in $\mathbb{R}^d$, then $\bigcap_{k=1}^n A_k$ is open in $\mathbb{R}^d$.

Next, let us defined closed sets by using open sets.

Definition 11.2.4 We say S is a closed set in $\mathbb{R}^d$ if $\mathbb{R}^d \setminus S$ is open.

Remark From the above definition, $\emptyset$ and $\mathbb{R}^d$ are both open and closed.

To study the structure of closed sets, let us firstly defined the boundary set.

Definition 11.2.5 We say P is a boundary point of S if $\forall r > 0$,

$$B_r(P) \cap S \neq \emptyset \quad \text{and} \quad B_r(P) \cap (\mathbb{R}^d \setminus S) \neq \emptyset.$$

The set of all boundary points is called the boundary of S . We denote it by ∂S . And $\overline{S} := S \cup \partial S$ is called the closure of S .

Example 11.2.6 $\partial(a, b) = \partial[a, b] = \{a, b\}$.

Note that if S is open in $\mathbb{R}^d$, then $S \cap \partial S = \emptyset$. And if S is closed in $\mathbb{R}^d$, then $\partial S \subseteq S$. Besides, we have the following result.

Proposition 11.2.7 Let $S \subseteq \mathbb{R}^d$. Then both ∂S and $\overline{S}$ are closed in $\mathbb{R}^d$.

Proof (1) $\forall P \in \mathbb{R}^d \setminus \partial S$, $\exists r > 0$ such that

$$B_r(P) \cap S = \emptyset \quad \text{or} \quad B_r(P) \cap (\mathbb{R}^d \setminus S) = \emptyset.$$

This implies every point in $B_r(P)$ can not be boundary points of S . Thus, $B_r(P) \subseteq \mathbb{R}^d \setminus \partial S$. Hence $\mathbb{R}^d \setminus \partial S$ is open, and ∂S is closed.

(2) $\forall P \in \mathbb{R}^d \setminus \overline{S}$, $P \notin \partial S$. So $\exists r > 0$ such that

$$B_r(P) \cap S = \emptyset \quad \text{or} \quad B_r(P) \cap (\mathbb{R}^d \setminus S) = \emptyset.$$

Since $x \notin S$, $B_r(P) \cap S = \emptyset$. This means $B_r(P) \subseteq \mathbb{R}^d \setminus S$ and $B_r(P) \cap \partial S = \emptyset$. Hence $\subseteq \mathbb{R}^d \setminus \overline{S}$ and $\mathbb{R}^d \setminus \overline{S}$ is open and $\overline{S}$ is closed. $\square$

Closed sets can be also described by using limit points.

Definition 11.2.8 We say P is a limit point of S if $\forall \varepsilon > 0$,

$$(B_\varepsilon(P) \setminus \{P\}) \cap S \neq \emptyset.$$

The set of all limit points is called the derived set of S . We denote it by S' .

Example 11.2.9 $B'_\varepsilon(P) = \overline{B}_\varepsilon(P)$.

Theorem 11.2.10 S is closed in $\mathbb{R}^d$ if and only if $S' \subseteq S$.

Proof On one hand, if S is closed, then $\mathbb{R}^d \setminus S$ is open. $\forall P \in \mathbb{R}^d \setminus S$, $\exists r > 0$ such that $B_r(P) \cap S = \emptyset$. This means $P \notin S'$. Hence $S' \subseteq S$.

On the other hand, if $S' \subseteq S$, then $\forall P \in \mathbb{R}^d \setminus S$, $P \notin S'$. Thus, $\exists r > 0$ such that

$$(B_r(P) \setminus \{P\}) \cap S = \emptyset.$$

Namely, $B_r(P) \subseteq \mathbb{R}^d \setminus S$. Hence $\mathbb{R}^d \setminus S$ is open, S is closed. $\square$

Theorem 11.2.11 Let $S \subseteq \mathbb{R}^d$. Then $\overline{S} = S \cup S'$.

Proof On one hand, since $\overline{S}$ is closed, $\overline{S'} \subseteq \overline{S}$. Thus,

$$S \cup S' \subseteq \overline{S} \cup \overline{S'} = \overline{S}.$$

On the other hand, $\forall P \in \partial S$, $B_\varepsilon(P) \cap S \neq \emptyset$ for all $\varepsilon > 0$. If $P \notin S$, then

$$(B_\varepsilon(P) \setminus \{P\}) \cap S \neq \emptyset, \quad \forall \varepsilon > 0.$$

This means $P \in S'$. Hence $\partial S \subseteq S \cup S'$. And it follows that $\overline{S} = S \cup \partial S \subseteq S \cup S'$.

Therefore, $\overline{S} = S \cup S'$. $\square$

Remark If S is a closed set in $\mathbb{R}^d$, then $\overline{S} = S$ and $\partial S \subseteq S$.

Finally, let us analyse the structure of open sets in $\mathbb{R}^d$.

Definition 11.2.12 Let $\vec{k} = (k_1, k_2, \dots, k_d) \in \mathbb{Z}^d$ and $m \in \mathbb{N}$. We call

$$Q_m(\vec{k}) = \left\{ \vec{x} = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d : \frac{k_j}{2^m} \leq x_j \leq \frac{k_j + 1}{2^m}, j = 1, 2, \dots, d \right\}.$$

the $\vec{k}$ -th dyadic cubes of order m . And

$$\mathcal{N}_m = \{Q_m(\vec{k}) : \vec{k} \in \mathbb{Z}^d\}$$

is called the dyadic net of order m in $\mathbb{R}^d$.

Remark It is easy to see dyadic cubes have the following properties:

1. $Q_m(\vec{k})$ are closed sets for all $m \in \mathbb{N}_+$ and $\vec{k} = (k_1, k_2, \dots, k_d) \in \mathbb{Z}^d$. And

$$\mathring{Q}_m(\vec{k}) = \left\{ \vec{x} = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d : \frac{k_j}{2^m} < x_j < \frac{k_j + 1}{2^m}, j = 1, 2, \dots, d \right\}$$

are all open cubes. The length of each side of $Q_m(\vec{k})$ is equal to 2^{-m} .

2. For each $m \in \mathbb{N}_+$, $\bigcup_{\vec{k} \in \mathbb{Z}^d} Q_m(\vec{k}) = \mathbb{R}^d$. And $\forall \vec{k}, \vec{k}' \in \mathbb{Z}^d$, if $\vec{k} \neq \vec{k}'$, then

$$\mathring{Q}_m(\vec{k}) \cap \mathring{Q}_m(\vec{k}') = \emptyset.$$

3. Every dyadic cube of order m can be rewritten as the union of 2^d dyadic cubes of order $m+1$. More specifically, $\forall \vec{k} = (k_1, k_2, \dots, k_d) \in \mathbb{Z}^d$, set

$$I(\vec{k}) = \left\{ \vec{\ell} = (\ell_1, \ell_2, \dots, \ell_d) \in \mathbb{Z}^d : \ell_j \in \{2k_j, 2k_j + 1\}, j = 1, 2, \dots, d \right\}.$$

Then $\text{card}(I(\vec{k})) = 2^d$. And $Q_m(\vec{k}) = \bigcup_{\vec{\ell} \in I(\vec{k})} Q_{m+1}(\vec{\ell})$.

Theorem 11.2.13 Let S be a nonempty open set in $\mathbb{R}^d$, then S can be written as the union of countably infinitely many different dyadic cubes.

Proof Let $\mathcal{S}_0$ be the set of all dyadic cubes of order 0 in S , and S_1 be the union of all dyadic cubes in $\mathcal{S}_0$.

Let $\mathcal{S}_1$ be the set of all dyadic cubes of order 1 in $S \setminus S_0$, and S_2 be the union of all dyadic cubes in $\mathcal{S}_1$.

Suppose we have already get $\mathcal{S}_k$ and S_k for $k = 0, 1, \dots, m$. Then let $\mathcal{S}_{m+1}$ be the set of all dyadic cubes of order $m+1$ in $S \setminus \left(\bigcup_{k=0}^m S_k\right)$, and S_{m+1} be the union of all dyadic cubes in $\mathcal{S}_{m+1}$.

Continuing this process, we get $\mathcal{S}_k$, $k \in \mathbb{N}$. Let

$$\mathcal{S} = \bigcup_{k=0}^{\infty} \mathcal{S}_k \quad \text{and} \quad \tilde{S} = \bigcup \{Q : Q \in \mathcal{S}\}.$$

It is easy to see $\tilde{S} \subseteq S$.

Since S is open, $\forall \vec{x} \in S$, $\exists r > 0$ such that $B_r(\vec{x}) \subseteq S$. Let $n \in \mathbb{N}_+$ be such that $2^{-n}\sqrt{d} < r$. Then there exists a dyadic cube Q_n of order n such that

$$\vec{x} \in Q_n \subsetneq B_r(\vec{x}) \subseteq S.$$

Thus, $\vec{x} \in \bigcup_{k=0}^n S_k \subseteq \tilde{S}$. This implies that $S \subseteq \tilde{S}$. And therefore, $S = \tilde{S}$. $\square$

11.2.2 Completeness and Compactness

Let us introduce the definition of limits of sequences in $\mathbb{R}^d$.

Definition 11.2.14 Suppose $\{P_n\}$ is a sequence of points in $\mathbb{R}^d$. And $P \in \mathbb{R}^d$. We say that $\{P_n\}$ converges to P as $n \rightarrow \infty$ and write $\lim_{n \rightarrow \infty} P_n = P$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}_+, \forall n > N, d(P_n, P) < \varepsilon.$$

It is easy to see the limit of a convergent sequence must be unique. And clearly, limits have linearity. The following result explains why closed sets are called closed.

Proposition 11.2.15 Let $\{P_n\}$ be a sequence of points in a closed set $D \subseteq \mathbb{R}^d$. If $\lim_{n \rightarrow \infty} P_n = P$, then $P \in D$.

Proof WLOG. we may assume that $P_n \neq P$ for all $n \in \mathbb{N}_+$. Since $\lim_{n \rightarrow \infty} P_n = P$, $\forall \varepsilon > 0, \exists N \in \mathbb{N}_+, \forall n > N, d(P_n, P) < \varepsilon$. Namely,

$$P_n \in B_\varepsilon(P), \quad \forall n > N.$$

Thus, $(B_\varepsilon(P) \setminus \{P\}) \cap D \neq \emptyset$ which means $P \in D'$. Since D is closed, $P \in D$. $\square$

Bolzano-Weierstrass Theorem also holds in $\mathbb{R}^d$.

Definition 11.2.16 We say a sequence $\{P_n\}$ is bounded in $\mathbb{R}^d$, if $\exists M > 0$ such that $\forall n \in \mathbb{N}_+, d(P_n, O) \leq M$.

Theorem 11.2.17 (Weierstrass-Bolzano) Every bounded sequence in $\mathbb{R}^d$ has a convergent subsequence.

Proof Suppose $\{P_n(x_n^{(1)}, x_n^{(2)}, \dots, x_n^{(d)})\}$ is a bounded sequence in $\mathbb{R}^d$. Then $\exists M > 0$, such that

$$d(P_n, O) = \left(\sum_{j=1}^d |x_n^{(j)}|^2 \right)^{\frac{1}{2}} \leq M.$$

It follows that $\forall j \in \{1, 2, \dots, d\}, \forall n \in \mathbb{N}_+, |x_n^{(j)}| \leq M$. Now, since $\{x_n^{(1)}\}$ is bounded, $\{x_n^{(1)}\}$ has a convergent subsequence $\{x_{n_{1k}}^{(1)}\}$.

Consider $\{P_{n_{1k}}(x_{n_{1k}}^{(1)}, x_{n_{1k}}^{(2)}, \dots, x_{n_{1k}}^{(d)})\}$. It is a subsequence of $\{P_n\}$. Since $\{x_{n_{1k}}^{(2)}\}$ is bounded, it has a convergent subsequence $\{x_{n_{2k}}^{(2)}\}$.

Since $\{P_{n_{2k}}(x_{n_{2k}}^{(1)}, x_{n_{2k}}^{(2)}, \dots, x_{n_{2k}}^{(d)})\} \subseteq \{P_{n_{1k}}\} \subseteq \{P_n\}$, $\{x_{n_{2k}}^{(1)}\}$ and $\{x_{n_{2k}}^{(2)}\}$ are both convergent. And since $\{x_{n_{2k}}^{(3)}\}$ is bounded, it has a convergent subsequence $\{x_{n_{3k}}^{(3)}\}$.

Continuing this process, we finally get $\{P_{n_{dk}}(x_{n_{dk}}^{(1)}, x_{n_{dk}}^{(2)}, \dots, x_{n_{dk}}^{(d)})\} \subseteq \{P_n\}$ such that $\{x_{n_{dk}}^{(1)}\}, \{x_{n_{dk}}^{(2)}\}, \dots, \{x_{n_{dk}}^{(d)}\}$ are all convergent.

Let $x^{(j)} = \lim_{k \rightarrow \infty} x_{n_{dk}}^{(j)}$ and $P(x^{(1)}, x^{(2)}, \dots, x^{(d)})$. Then $\forall \varepsilon > 0, \exists N_j \in \mathbb{N}_+$ such that

$$|x_{n_{dk}}^{(j)} - x^{(j)}| < \frac{\varepsilon}{\sqrt{d}}, \quad \forall k > N_j.$$

Set $N = \max\{N_1, N_2, \dots, N_d\}$, then $\forall k > N$,

$$d(P_{n_{dk}}, P) = \left(\sum_{j=1}^d (x_{n_{dk}}^{(j)} - x^{(j)})^2 \right)^{\frac{1}{2}} < \varepsilon.$$

Therefore, $\lim_{n \rightarrow \infty} P_{n_{dk}} = P$. □

In $\mathbb{R}^1$, we used Monotone Convergent Theorem to describe the completeness of real line. But the monotonicity of a high dimensional points is very difficult to define. However, we know that Cauchy Criterion and Monotone Convergent Theorem are equivalent. So let us use Cauchy Criterion to describe the completeness of $\mathbb{R}^d$.

Definition 11.2.18 We say $\{P_n\}$ is a Cauchy sequence in $\mathbb{R}^d$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}_+, \forall n, m > N, d(P_n, P_m) < \varepsilon.$$

Theorem 11.2.19 (Cauchy Criterion) A sequence in $\mathbb{R}^d$ is convergent if and only if it is a Cauchy sequence.

Proof On one hand, suppose $\{P_n\}$ converges to P in $\mathbb{R}^d$. Then $\forall \varepsilon > 0, \exists N \in \mathbb{N}_+$ such that $\forall n > N, d(P_n, P) < \frac{\varepsilon}{2}$. Hence, $\forall m, n > N$,

$$d(P_n, P_m) \leq d(P_n, P) + d(P_m, P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This means that $\{P_n\}$ is a Cauchy sequence.

On the other hand, suppose $\{P_n(x_n^{(1)}, x_n^{(2)}, \dots, x_n^{(d)})\}$ is a Cauchy sequence. Then $\forall \varepsilon > 0, \exists N \in \mathbb{N}_+$ such that $\forall m, n > N, d(P_n, P_m) < \varepsilon$. It follows that $\forall j \in \{1, 2, \dots, d\}$,

$$|x_n^{(j)} - x_m^{(j)}| \leq \left(\sum_{k=1}^d (x_n^{(k)} - x_m^{(k)})^2 \right)^{\frac{1}{2}} = d(P_n, P_m) < \varepsilon.$$

Hence $\{x_n^{(j)}\}$ ($j = 1, 2, \dots, d$) are all Cauchy sequences. By Cauchy Criterion, $\{x_n^{(j)}\}$ is convergent for all $j = 1, 2, \dots, d$. Let $x^{(j)} = \lim_{n \rightarrow \infty} x_n^{(j)}$ and $P(x^{(1)}, x^{(2)}, \dots, x^{(d)})$. Then $\exists N_j \in \mathbb{N}_+$ such that $\forall n > N_j$,

$$|x_n^{(j)} - x^{(j)}| < \frac{\varepsilon}{\sqrt{d}}.$$

Set $N = \max\{N_1, N_2, \dots, N_d\}$, then $\forall n > N$,

$$d(P_n, P) = \left(\sum_{j=1}^d (x_n^{(j)} - x^{(j)})^2 \right)^{\frac{1}{2}} < \varepsilon.$$

Therefore, $\lim_{n \rightarrow \infty} P_n = P$. □

From this result, we know that $\mathbb{R}^d$ is a complete Hilbert space. The completeness of $\mathbb{R}^d$ can be also described by some other ways.

Definition 11.2.20 Suppose $S \subseteq \mathbb{R}^d$. $\text{diam}(S) := \sup\{d(P, Q) : P, Q \in S\}$.

Theorem 11.2.21 (Closed Nested Sets) Suppose $\{D_n\}$ is a sequence of nonempty closed sets in $\mathbb{R}^d$. If $D_{n+1} \subseteq D_n$ and $\lim_{n \rightarrow \infty} \text{diam}(D_n) = 0$, then there exists unique $P \in \mathbb{R}^d$ such that $P \in \bigcap_{n=1}^{\infty} D_n$.

Proof Since $\lim_{n \rightarrow \infty} \text{diam}(D_n) = 0, \forall \varepsilon > 0, \exists N \in \mathbb{N}_+, \forall n > N, \text{diam}(D_n) < \varepsilon$.

Let $P_n \in D_n$. Then $\forall m > n > N, P_m \in D_m \subseteq D_n$. And it follows that

$$d(P_m, P_n) \leq \text{diam}(D_n) < \varepsilon.$$

This means $\{P_n\}$ is a Cauchy sequence. Set $P = \lim_{n \rightarrow \infty} P_n$.

$\forall k \in \mathbb{N}_+, P_n \in D_k$ for all $n \geq k$. Since D_k is closed, $P \in D_k$. Thus $P \in \bigcap_{n=1}^{\infty} D_n$.

Suppose $Q \in \bigcap_{n=1}^{\infty} D_n$, then $d(P, Q) \leq \text{diam}(D_n)$ for all $n \in \mathbb{N}_+$. Since $\lim_{n \rightarrow \infty} \text{diam}(D_n) = 0, d(P, Q) = 0$. This means $P = Q$. □

Definition 11.2.22 Let I be an index set. We say a family of open sets $\mathcal{O} = \{U_\alpha : \alpha \in I\}$ is an open covering of $S \subseteq \mathbb{R}^d$ if

$$S \subseteq \bigcup_{\alpha \in I} U_\alpha.$$

And if $\exists U_1, U_2, \dots, U_n \in \mathcal{O}$ such that $S \subseteq \bigcup_{k=1}^n U_k$, then $\{U_k : k = 1, 2, \dots, n\}$ is called a finite subcovering of S .

Given a set $S \subseteq \mathbb{R}^d$. We say S is compact if every open covering of S has a finite subcovering.

Definition 11.2.23 We say $S \subseteq \mathbb{R}^d$ is bounded if $\exists M > 0$ such that $S \subseteq B_M(O)$.

Theorem 11.2.24 A set in $\mathbb{R}^d$ is compact if and only if it is bounded and closed.

Proof On the one hand, let S be a bounded closed set in $\mathbb{R}^d$. Then S is contained in a closed cube Q in $\mathbb{R}^d$.

Suppose $\mathcal{O} = \{U_\alpha : \alpha \in I\}$ is an open covering of S , but S cannot be covered by any finitely many subsets of $\mathcal{O}$. Now, separate Q into 2^d equal closed cubes $Q_j^{(1)}$ ($j = 1, 2, \dots, 2^d$). And set $S_j^{(1)} = S \cap Q_j^{(1)}$. There is at least one of $S_j^{(1)}$ ($j = 1, 2, \dots, 2^d$) cannot be covered by finitely many subsets of $\mathcal{O}$. We denote it by S_1 , and suppose $S_1 \subseteq Q_1$.

Then separate Q_1 into 2^d equal closed cubes $Q_j^{(2)}$ ($j = 1, 2, \dots, 2^d$). And set $S_j^{(2)} = S_1 \cap Q_j^{(2)}$. There is at least one of $S_j^{(2)}$ ($j = 1, 2, \dots, 2^d$) cannot be covered by finitely many subsets of $\mathcal{O}$. We denote it by S_2 .

Continuing this process, we get a sequence nonempty closed sets $\{S_n\}$ with

$$\lim_{n \rightarrow \infty} \text{diam}(S_n) = 0.$$

By Theorem of Closed Nested Sets, $\exists P \in \bigcap_{n=1}^{\infty} S_n$.

Since S is closed, $P \in S$. So $\exists \alpha_0 \in I$ such that $P \in U_{\alpha_0}$. Since U_{α_0} is open, $\exists \delta > 0$ such that $P \in B_\delta(P) \subseteq U_{\alpha_0}$. Since $\lim_{n \rightarrow \infty} \text{diam}(S_n) = 0$, $\exists N \in \mathbb{N}_+$ such that

$$P \in S_N \subseteq B_\delta(P) \subseteq U_{\alpha_0}.$$

This is a contradiction since we once assumed that S_N cannot be covered by finitely many subsets of $\mathcal{O}$.

On the other hand, let K be a nonempty compact set in $\mathbb{R}^d$. Then $\mathcal{S} = \{B_n(O) : n \in \mathbb{N}_+\}$ must be an open cover covering of K . If K is unbounded, then any finite subset of $\mathcal{S}$ can not cover K . Contradiction! So K is bounded.

Let $\vec{x} \in \mathbb{R}^d \setminus K$. $\forall \vec{y} \in K$, $\delta_y = 2^{-1}d(\vec{x}, \vec{y}) > 0$. Since

$$K \subseteq \bigcup_{\vec{y} \in K} B_{\delta_y}(\vec{y}),$$

there exists $\vec{y}_1, \vec{y}_2, \dots, \vec{y}_n \in K$ such that

$$K \subseteq \bigcup_{j=1}^n B_{\delta_j}(\vec{y}_j),$$

where $\delta_j = 2^{-1}d(\vec{x}, \vec{y}_j)$ ($j = 1, 2, \dots, n$).

Let $\delta = \min\{\delta_1, \delta_2, \dots, \delta_n\}$, then $B_\delta(\vec{x}) \cap K = \emptyset$. This means $B_\delta(\vec{x}) \subseteq \mathbb{R}^d \setminus K$. Hence $\mathbb{R}^d \setminus K$ is open and K is closed. $\square$

11.3 Lebesgue Measure on $\mathbb{R}^d$

In primary and middle school, we learned lengths of segments in $\mathbb{R}^1$, the areas of discs and polygons in $\mathbb{R}^2$, and the volume of balls and polyhedrons in $\mathbb{R}^3$. Now, I want to ask you a very nature question. What is the name of the analog of length, area and volume on $\mathbb{R}^d$ when $d \geq 4$? It is called measure. And length, area, volume are also measures on $\mathbb{R}^1$, $\mathbb{R}^2$ and $\mathbb{R}^3$.

The measure is an abstract concept. In this section, we are devoted to give the mathematical definition of measure and study the properties of measure. To do this work, we need to strictly follow our intuition, summarize the common characters of length, area, volume. And avoid making any contradictions.

11.3.1 Outer Measure

Let $\vec{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^d$ and $\ell_1, \ell_2, \dots, \ell_d \geq 0$. The set

$$Q = \{(x_1, x_2, \dots, x_d) \in \mathbb{R}^d : a_j < x_j < a_j + \ell_j, j = 1, 2, \dots, d\}.$$

is called an open cuboid. The volume of Q is denoted by $|Q| = \ell_1 \ell_2 \cdots \ell_d$.

Let $B \subseteq \partial Q$, then $B \cup Q$ is called a (general) cuboid. And if $B = \partial Q$, then $B \cup Q$ is called a closed cuboid. The center of $B \cup Q$ is $\vec{a} + 2^{-1}\vec{\ell}$, where $\vec{\ell} = (\ell_1, \ell_2, \dots, \ell_d)$.

Definition 11.3.1 Let $E \subset \mathbb{R}^d$. The (Lebesgue) outer measure of E is

$$m^*(E) = \inf \left\{ \sum_{k=1}^{\infty} |Q_k| : Q_k \text{ are all open cuboids and } E \subseteq \bigcup_{k=1}^{\infty} Q_k \right\}.$$

Remark 1. From the definition of outer measure, $m^*(E) \geq 0$ for all $E \subseteq \mathbb{R}^d$.

2. $\emptyset$ can be also regarded as a open cube with side length 0. Under the case when E is covered by finitely many nonempty open cuboids $Q_1, Q_2, \dots, Q_n$, we set $Q_k = \emptyset$ ($k > n$). Then

$$E \subseteq \bigcup_{k=1}^n Q_k = \bigcup_{k=1}^{\infty} Q_k.$$

Let us firstly evaluate the outer measures of some specific sets.

Example 11.3.2 $m^*(\emptyset) = 0$.

Proof Set $Q_n = \emptyset$ for all $n \in \mathbb{N}_+$. Then $\emptyset \subseteq \bigcup_{k=1}^{\infty} Q_k$ and

$$0 \leq m^*(\emptyset) \leq \sum_{k=1}^{\infty} |Q_k| = 0.$$

Thus, $m^*(\emptyset) = 0$. □

Example 11.3.3 If E is a set of single point in $\mathbb{R}^d$, then $m^*(E) = 0$.

Proof Let $E = \{\vec{a}\}$. $\forall \varepsilon > 0$, let $Q(\varepsilon)$ be an open cube centered at $\vec{a}$ with side length ε . Then $Q(\varepsilon)$ is a special open covering of E . Thus,

$$m^*(E) = \inf \left\{ \sum_{k=1}^{\infty} |Q_k| : Q_k \text{ are all open cuboids and } E \subseteq \bigcup_{k=1}^{\infty} Q_k \right\} \leq |Q(\varepsilon)| = \varepsilon^d.$$

Since $m^*(E) \geq 0$, we get $m^*(E) = 0$ by letting $\varepsilon \rightarrow 0$. □

The outer measure has following properties.

Proposition 11.3.4 (Monotonicity) If $A \subset B \subseteq \mathbb{R}^d$, then $m^*(A) \leq m^*(B)$.

Proof Let $\mathcal{S}_A = \left\{ \{Q_n\} : Q_k \text{ are all open cuboids and } A \subseteq \bigcup_{k=1}^{\infty} Q_k \right\}$ and $\mathcal{S}_B = \left\{ \{Q_n\} : Q_k \text{ are all open cuboids and } B \subseteq \bigcup_{k=1}^{\infty} Q_k \right\}$. Since $A \subset B$, $\mathcal{S}_B \subseteq \mathcal{S}_A$. Thus,

$$m^*(A) = \inf \left\{ \sum_{k=1}^{\infty} |Q_k| : \{Q_n\} \in \mathcal{S}_A \right\} \leq \inf \left\{ \sum_{k=1}^{\infty} |Q_k| : \{Q_k\} \in \mathcal{S}_B \right\} = m^*(B).$$

□

This property can help us evaluate the outer measure of open cuboids and closed cuboids.

Example 11.3.5 Let Q be a bounded open cuboids in $\mathbb{R}^d$. Then

$$m^*(Q) = m^*(\overline{Q}) = |Q|.$$

Proof Let $Q = \{(x_1, x_2, \dots, x_d) : a_j < x_j < a_j + \ell_j, j = 1, 2, \dots, d\}$.

One the one hand, $\forall \varepsilon \in (0, 2^{-1} \min\{\ell_1, \ell_2, \dots, \ell_d\})$, set

$$F = \{(x_1, x_2, \dots, x_d) : a_j + \varepsilon \leq x_j \leq a_j + \ell_j - \varepsilon, j = 1, 2, \dots, d\}.$$

Then F is a closed cuboid contained in Q . If $\{Q_n\}$ is a sequence open cuboids that covers Q , then $\{Q_n\}$ is an open covering of F . So there exists finite cuboids, say $Q_1, \dots, Q_N$, that covers F . Clearly,

$$\sum_{k=1}^{\infty} |Q_k| \geq \sum_{k=1}^N |Q_k| \geq |\overset{\circ}{F}| = \prod_{j=1}^d (\ell_j - 2\varepsilon).$$

It follows that $m^*(Q) \geq \prod_{j=1}^d (\ell_j - 2\varepsilon)$. Letting $\varepsilon \rightarrow 0$, we get $m^*(Q) \geq \prod_{j=1}^d \ell_j = |Q|$.

On the other hand, $\forall \varepsilon > 0$, let

$$\tilde{Q} = \{(x_1, x_2, \dots, x_d) : a_j - \varepsilon < x_j < a_j + \ell_j + \varepsilon, j = 1, 2, \dots, d\}.$$

Then $\tilde{Q}$ is an open cuboid containing $\overline{Q}$. So $m^*(\overline{Q}) \leq |\tilde{Q}| = \prod_{j=1}^d (\ell_j + 2\varepsilon)$. Letting

$\varepsilon \rightarrow 0$, we get $m^*(\overline{Q}) \leq \prod_{j=1}^d \ell_j = |Q|$.

Since $|Q| \leq m^*(Q) \leq m^*(\overline{Q}) \leq |Q|$, $m^*(Q) = m^*(\overline{Q}) = |Q|$. $\square$

Remark For a general cuboid Q which may just contain part of its boundary, $\mathring{Q}$ is an open cuboid and $\overline{Q}$ is a closed cuboid. Since

$$\mathring{Q} \subseteq Q \subseteq \overline{Q},$$

the monotonicity of outer measure gives that $m^*(\mathring{Q}) = m^*(Q) = m^*(\overline{Q}) = |\mathring{Q}|$.

Proposition 11.3.6 (Translation Invariance) $\forall E \subseteq \mathbb{R}^d, \vec{x}_0 \in \mathbb{R}^d$. Define

$$\vec{x}_0 + E = E + \vec{x}_0 = \{\vec{x} + \vec{x}_0 : \vec{x} \in E\}.$$

Then $m^*(\vec{x}_0 + E) = m^*(E)$.

Proof If Q is a open cuboid in $\mathbb{R}^d$, then $\vec{x}_0 + Q$ is also a open cuboid in $\mathbb{R}^d$.

Let $\mathcal{S} = \left\{ \{Q_n\} : Q_k \text{ are all open cuboids and } E \subseteq \bigcup_{k=1}^{\infty} Q_k \right\}$. Since

$$\vec{x}_0 + E \subseteq \bigcup_{k=1}^{\infty} (\vec{x}_0 + Q_k), \quad \forall \{Q_k\} \in \mathcal{S},$$

we get that

$$m^*(\vec{x}_0 + E) \leq \sum_{k=1}^{\infty} |\vec{x}_0 + Q_k| = \sum_{k=1}^{\infty} |Q_k|, \quad \forall \{Q_k\} \in \mathcal{S}.$$

Then, it follows that

$$m^*(\vec{x}_0 + E) \leq \inf \left\{ \sum_{k=1}^{\infty} |Q_k| : \{Q_k\} \in \mathcal{S} \right\} = m^*(E).$$

However, $m^*(E) = m^*(-\vec{x}_0 + (\vec{x}_0 + E))$. So $m^*(E) \leq m^*(\vec{x}_0 + E)$. Therefore, $m^*(E) = m^*(\vec{x}_0 + E)$. $\square$

Proposition 11.3.7 (Subadditivity) Let $E_n \subseteq \mathbb{R}^d$ and $E = \bigcup_{n=1}^{\infty} E_n$, then

$$m^*(E) \leq \sum_{n=1}^{\infty} m^*(E_n).$$

Proof If $\exists E_k$ such that $m^*(E_k) = \infty$, then the inequality holds.

Suppose $m^*(E_n) < +\infty$ for all $n \in \mathbb{N}_+$. Then $\forall \varepsilon > 0$, there exists a sequence of open cuboids $\{Q_{nj}\}_{j=1}^{\infty}$ such that

$$E_n \subseteq \bigcup_{j=1}^{\infty} Q_{nj} \quad \text{and} \quad \sum_{j=1}^{\infty} |Q_{nj}| < m^*(E_n) + \frac{\varepsilon}{2^n}.$$

It follows that

$$E = \bigcup_{n=1}^{\infty} E_n \subseteq \bigcup_{n=1}^{\infty} \bigcup_{j=1}^{\infty} Q_{nj} \quad \text{and} \quad \sum_{n=1}^{\infty} \sum_{j=1}^{\infty} |Q_{nj}| < \sum_{n=1}^{\infty} m^*(E_n) + \varepsilon.$$

Then, by definition

$$m^*(E) < \sum_{k=1}^{\infty} m^*(E_k) + \varepsilon.$$

Letting $\varepsilon \rightarrow 0$, we can get the desired inequality. $\square$

Remark 1. Subadditivity also holds under the case of finite union. If $E = \bigcup_{n=1}^N E_n$, we set $E_k = \emptyset$ for $k \geq N + 1$. Then

$$m^*\left(\bigcup_{n=1}^N E_n\right) = m^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m^*(E_n) \leq \sum_{n=1}^N m^*(E_n).$$

2. By the subadditivity and the result of Example 11.3.3, the outer measures of finite and countable sets are all zero.

At last, we would like to show that outer measure has finite additivity for sets with positive distance.

Proposition 11.3.8 For nonempty sets $A, B \subset \mathbb{R}^d$, the distance between A and B is defined as

$$d(A, B) = \inf \{ \|\vec{x} - \vec{y}\| : \vec{x} \in A, \vec{y} \in B \}.$$

If $d(A, B) > 0$, then $m^*(A \cup B) = m^*(A) + m^*(B)$.

Proof $\forall \varepsilon > 0$, there exists a sequence of open cuboids $\{Q_n\}$ such that

$$A \cup B \subseteq \bigcup_{n=1}^{\infty} Q_n \quad \text{and} \quad \sum_{n=1}^{\infty} |Q_n| < m^*(A \cup B) + \varepsilon.$$

Divide each Q_n into several disjoint smaller open cuboids Q_{nk} such that the length of all sides of Q_{nk} are less than $(2\sqrt{d})^{-1}d(A, B)$ and $|Q_n| = \sum_k |Q_{nk}|$.

Let $\vec{x}_{nk}$ be the center of Q_{nk} . For each $\lambda \in (1, 2)$, let λQ_{nk} be open cuboids centered at $\vec{x}_{nk}$ and with side-length λ times as much as Q_{nk} . Then $|\lambda Q_{nk}| = \lambda^d |Q_{nk}|$.

1. For each fixed $n \in \mathbb{N}_+$, Q_n is covered by the union of all λQ_{nk} ,
2. The length of the sides open cuboids λQ_{nk} are all less than $(\sqrt{d})^{-1}d(A, B)$.

Let $\mathcal{S}_A = \{\lambda Q_{nk} : \lambda Q_{nk} \cap A \neq \emptyset\}$ and $\mathcal{S}_B = \{\lambda Q_{nk} : \lambda Q_{nk} \cap B \neq \emptyset\}$. Then the open cuboids in $\mathcal{S}_A$, $\mathcal{S}_B$ covers A , B respectively. And $\mathcal{S}_A \cap \mathcal{S}_B = \emptyset$. Hence

$$\begin{aligned} m^*(A) + m^*(B) &\leq \sum_{Q \in \mathcal{S}_A} |Q| + \sum_{Q \in \mathcal{S}_B} |Q| \\ &\leq \sum_{n=1}^{\infty} \sum_k |\lambda Q_{nk}| = \lambda^d \sum_{n=1}^{\infty} \sum_k |Q_{nk}| = \lambda^d \sum_{n=1}^{\infty} |Q_n| \\ &\leq \lambda^d m^*(A \cup B) + \lambda^d \varepsilon. \end{aligned}$$

Letting $\lambda \rightarrow 1^+$, we get that $m^*(A) + m^*(B) \leq m^*(A \cup B) + \varepsilon$. Then letting $\varepsilon \rightarrow 0$, we get the $m^*(A) + m^*(B) \leq m^*(A \cup B)$. This implies that

$$m^*(A) + m^*(B) = m^*(A \cup B).$$

□

Example 11.3.9 Let $\{Q_n\}$ be a sequence of cuboids (may not be open or closed) in $\mathbb{R}^d$. If $\{\overset{\circ}{Q}_n\}$ are disjoint, then

$$m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) = \sum_{n=1}^{\infty} m^*(Q_n).$$

Proof Let $\vec{x}_n$ be the center of Q_n . $\forall \lambda \in (0, 1)$, set $Q_n(\lambda)$ be open cuboids centered at $\vec{x}_n$ and with side-length λ times as much as Q_n . Then $d(Q_n(\lambda), Q_m(\lambda)) > 0$ if $m \neq n$. By monotonicity and Proposition 11.3.8, $\forall N \in \mathbb{N}_+$

$$\begin{aligned} m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) &\geq m^*\left(\bigcup_{n=1}^N Q_n\right) \geq m^*\left(\bigcup_{n=1}^N Q_n(\lambda)\right) = \sum_{n=1}^N m^*(Q_n(\lambda)) \\ &= \sum_{n=1}^N |Q_n(\lambda)| = \lambda^d \sum_{n=1}^N |Q_n| = \lambda^d \sum_{n=1}^N m^*(Q_n). \end{aligned}$$

Letting $\lambda \rightarrow 1^-$ and $N \rightarrow \infty$, we get $m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) \geq \sum_{n=1}^{\infty} m^*(Q_n)$. By subadditivity,

$$m^*\left(\bigcup_{n=1}^{\infty} Q_n\right) = \sum_{n=1}^{\infty} m^*(Q_n).$$

□

11.3.2 Measurable Sets

From last subsection, we may see that outer measure is consistent with almost all of intuitive feeling about the length, area or volume that we learned in high school. So a very natural question is whether outer measure could become the generalization of them on $\mathbb{R}^d$. But the answer to this question is No.

In fact, to generalize the length on $\mathbb{R}^d$, we need to find a mapping

$$\mu : \{\text{all subsets of } \mathbb{R}^d\} \rightarrow \mathbb{R}_+ \cup \{0, +\infty\}$$

which should satisfy at least the following five properties:

1. If $A \subseteq B \subseteq \mathbb{R}^d$, then $\mu(A) \leq \mu(B)$;
2. For each cuboid $Q \subseteq \mathbb{R}^d$, $\mu(Q) = |Q|$;
3. For all $\vec{x}_0 \in \mathbb{R}^d$ and $S \subseteq \mathbb{R}^d$, $\mu(\vec{x}_0 + S) = \mu(S)$;
4. If $\{S_n\}$ is a sequence of subsets of $\mathbb{R}^d$, then $\mu\left(\bigcup_{n=1}^{\infty} S_n\right) \leq \sum_{n=1}^{\infty} \mu(S_n)$;
5. For each sequence of disjoint subsets $\{S_n\} \subseteq \mathbb{R}^d$, $\mu\left(\bigcup_{n=1}^{\infty} S_n\right) = \sum_{n=1}^{\infty} \mu(S_n)$.

But unfortunately, this mapping μ actually does not exist even when $d = 1$. Let us explain the reason as follows.

Suppose μ exists! $\forall x \in (0, 1)$, set $E_x = \{\xi \in (0, 1) : \xi - x \in \mathbb{Q}\}$. Since $x \in E_x$, $E_x \neq \emptyset$ for all $x \in (0, 1)$. Let S be a set in $(0, 1)$ such that

$$\forall x \in (0, 1), S \text{ contains exactly one element in } E_x.$$

Choose $\{q_n\} = (-1, 1) \cap \mathbb{Q}$, and let $S_n = q_n + S$. Then $S_n \subseteq (-1, 2)$ for all $n \in \mathbb{N}_+$.

If $S_n \cap S_m \neq \emptyset$, then $\exists x, y \in S$ such that

$$q_n + x = q_m + y.$$

Thus, $x - y = q_m - q_n \in \mathbb{Q}$. This means $x \in E_y$. By the principle of the construction of S , $x = y$. Hence $q_n = q_m$ and $n = m$. Therefore, $\{S_n\}$ are disjoint.

Since $S_n \subseteq (-1, 2)$, $\bigcup_{n=1}^{\infty} S_n \subseteq (-1, 2)$. We claim that $(0, 1) \subseteq \bigcup_{n=1}^{\infty} S_n$.

In fact, $\forall x \in (0, 1)$, $\exists \alpha \in E_x \cap S$ such that $x - \alpha \in (-1, 1) \cap \mathbb{Q}$. So $\exists k \in \mathbb{N}_+$ such that $x - \alpha = q_k$, which means

$$x = q_k + \alpha \in S_k \subseteq \bigcup_{n=1}^{\infty} S_n.$$

Hence $(0, 1) \subseteq \bigcup_{n=1}^{\infty} S_n \subseteq (-1, 2)$. By Property 1 and 2, we get $1 \leq \mu\left(\bigcup_{n=1}^{\infty} S_n\right) \leq 3$.

However, by Property 3, $\mu(S_n) = \mu(S)$ for all $n \in \mathbb{N}_+$. If $\mu(S) = 0$,

$$\sum_{n=1}^{\infty} \mu(S_n) = 0 < 1 = \mu\left(\bigcup_{n=1}^{\infty} S_n\right).$$

This contradicts to Property 4! But if $\mu(S) > 0$, $\sum_{n=1}^{\infty} \mu(S_n) = \sum_{n=1}^{\infty} \mu(S) = +\infty$. So

$$\mu\left(\bigcup_{n=1}^{\infty} S_n\right) < \sum_{n=1}^{\infty} \mu(S_n),$$

which contradicts to Property 5! This example shows that none of the mapping

$$\mu : \{\text{all subsets of } \mathbb{R}^d\} \rightarrow \mathbb{R}_+ \cup \{0, +\infty\}$$

could satisfies the above five properties simultaneously.

Therefore, to defined a measure consisting with intuitive perception, we have to narrow the domain of mapping μ and accept the existence of nonmeasurable sets.

Next, our work is to screen out all measurable sets in $\mathbb{R}^d$ based on the outer measure m^* we defined in the last subsection. Firstly, note that all open cuboids must be measurable. So if a set E is measurable in $\mathbb{R}^d$, then it should at least satisfy the additivity property with all open cuboids. Namely, for all open cuboids $Q \in \mathbb{R}^d$,

$$m^*(Q) = m^*(Q \cap E) + m^*(Q \cap E^c). \quad (11.1)$$

However, these condition actually implies that $m^*(T) = m^*(T \cap E) + m^*(T \cap E^c)$ for all $T \in \mathbb{R}^d$.

In fact, $\forall \varepsilon > 0$, there exists a sequence of open cuboids $\{Q_n\}$ such that

$$T \subseteq \bigcup_{n=1}^{\infty} Q_k \quad \text{and} \quad \sum_{n=1}^{\infty} |Q_k| < m^*(T) + \varepsilon.$$

It follows that

$$\begin{aligned} (E \cap T) &\subseteq E \cap \left(\bigcup_{n=1}^{\infty} Q_k\right) = \bigcup_{n=1}^{\infty} (E \cap Q_k), \\ (E^c \cap T) &\subseteq E^c \cap \left(\bigcup_{n=1}^{\infty} Q_k\right) = \bigcup_{n=1}^{\infty} (E^c \cap Q_k). \end{aligned}$$

Then by (11.1),

$$\begin{aligned} m^*(T \cap E) + m^*(T \cap E^c) &\leq m^*\left(\bigcup_{n=1}^{\infty} (Q_k \cap E)\right) + m^*\left(\bigcup_{n=1}^{\infty} (Q_k \cap E^c)\right) \\ &\leq \sum_{n=1}^{\infty} [m^*(Q_k \cap E) + m^*(Q_k \cap E^c)] \\ &= \sum_{n=1}^{\infty} m^*(Q_k) = \sum_{n=1}^{\infty} |Q_k| \leq m^*(T) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \rightarrow 0$, we get $m^*(T \cap E) + m^*(T \cap E^c) \leq m^*(T)$.

The inverse inequality just follows that fact $T = (T \cap E) \cup (T \cap E^c)$ and the subadditivity of outer measure.

We shall use this equation to screen out measurable sets in $\mathbb{R}^d$.

Definition 11.3.10 We say a set $E \subset \mathbb{R}^d$ is (Lebesgue) measurable if

$$m^*(T) = m^*(T \cap E) + m^*(T \cap E^c), \quad \forall T \in \mathbb{R}^d. \quad (11.2)$$

If E is measurable, then the measure of E , say $m(E)$ or $|E|$, is defined as $m^*(E)$. And we denote the collection of all measurable sets in $\mathbb{R}^d$ by $\mathcal{M}_d$.

Remark 1. Equation (11.2) is call Carathéodory condition, since it was given by Greek mathematician Constantin Carathéodory (1873 - 1950).

2. Since $T = (T \cap E) \cup (T \cap E^c)$, the subadditivity of outer measure directly gives that $m^*(T) \leq m^*(T \cap E) + m^*(T \cap E^c)$. So to show a set E is measurable, we just need to show $\forall T \in \mathbb{R}^d$,

$$m^*(T) \geq m^*(T \cap E) + m^*(T \cap E^c). \quad (11.3)$$

Or showing inequality (11.3) for all open cuboids.

Example 11.3.11 $\emptyset, \mathbb{R}^d \in \mathcal{M}_d$. And $E \in \mathcal{M}_d$ if $m^*(E) = 0$.

Proof $\forall T \subseteq \mathbb{R}^d$, $T \cap \emptyset = T \cap (\mathbb{R}^d)^c = \emptyset$ and $T \cap \emptyset^c = T \cap \mathbb{R}^d = T$. Thus,

$$m^*(T) = m^*(T \cap \emptyset) + m^*(T \cap \emptyset^c) = m^*(T \cap \mathbb{R}^d) + m^*(T \cap (\mathbb{R}^d)^c).$$

This means that $\emptyset$ and $\mathbb{R}^d$ are both measurable.

$\forall T \subseteq \mathbb{R}^d$, $m^*(T \cap E) \leq m^*(E)$. If $m^*(E) = 0$, then $m^*(T \cap E) = 0$ for all $T \subseteq \mathbb{R}^d$. Since $T = (T \cap E) \cup (T \cap E^c)$,

$$m^*(T) \leq m^*(T \cap E) + m^*(T \cap E^c) = m^*(T \cap E^c).$$

But the monotonicity gives $m^*(T) \geq m^*(T \cap E^c)$. So $m^*(T) = m^*(T \cap E^c)$. Hence

$$m^*(T) = m^*(T \cap E) + m^*(T \cap E^c),$$

which means that E is measurable. □

Example 11.3.12 For each cuboid $Q \subseteq \mathbb{R}^d$, $Q \in \mathcal{M}_d$. And $m(Q) = |Q|$.

Proof Let $\tilde{Q}$ be an arbitrary cuboid in $\mathbb{R}^d$. Then $\tilde{Q} \cap Q$ is a still a cuboid, and $\tilde{Q} \cap Q^c$ can be decomposed into finitely many cuboids $Q_1, Q_2, \dots, Q_N$ such that

$$\bigcup_{j=1}^N Q_j = \tilde{Q} \cap Q^c \quad \text{and} \quad \tilde{Q}_i \cap \tilde{Q}_j = \emptyset \quad (1 \leq i < j \leq N).$$

Let $Q_0 = \tilde{Q} \cap Q$, then $\tilde{Q} = \bigcup_{j=0}^N Q_j$ and $\{\tilde{Q}_j : j = 0, 1, \dots, N\}$ are disjoint. By the result of Example 11.3.9,

$$\begin{aligned} m^*(\tilde{Q}) &= m^*\left(\bigcup_{j=0}^N Q_j\right) = m^*(Q_0) + \sum_{j=1}^N m^*(Q_j) = m^*(\tilde{Q} \cap Q) + m^*\left(\bigcup_{j=1}^N Q_j\right) \\ &= m^*(\tilde{Q} \cap Q) + m^*(\tilde{Q} \cap Q^c). \end{aligned}$$

Therefore, $Q \in \mathcal{M}_d$ and $m(Q) = m^*(Q) = |Q|$. $\square$

Next, we will introduce some properties of Lebesgue measures. These properties will help us verify the measurability of more general sets.

Proposition 11.3.13 If $E, F \in \mathcal{M}_d$, then $E \setminus F \in \mathcal{M}_d$.

Proof For an arbitrary set $T \subseteq \mathbb{R}^d$,

$$\begin{aligned} m^*(T) &= m^*(T \cap E^c) + m^*(T \cap E) \\ &= m^*(T \cap E^c) + m^*(T \cap E \cap F) + m^*(T \cap E \cap F^c). \end{aligned}$$

Since $(T \cap E^c) \cup (T \cap E \cap F) = T \cap (E^c \cup (E \cap F)) = T \cap (E^c \cup F)$,

$$m^*(T \cap E^c) + m^*(T \cap E \cap F) \geq m^*(T \cap (E^c \cup F)).$$

Thus,

$$\begin{aligned} m^*(T) &\geq m^*(T \cap (E^c \cup F)) + m^*(T \cap E \cap F^c) \\ &= m^*(T \cap (E \cap F^c)^c) + m^*(T \cap (E \cap F^c)). \end{aligned}$$

This means $E \cap F^c = E \setminus F \in \mathcal{M}_d$. $\square$

Remark This property tells us $E^c = \mathbb{R}^d \setminus E \in \mathcal{M}_d$ if $E \in \mathcal{M}_d$.

Proposition 11.3.14 (Additivity) Let $\{E_n\}$ be a sequence of disjoint measurable sets contained in $\mathbb{R}^d$. Then $E = \bigcup_{n=1}^{\infty} E_n$ is measurable and

$$m(E) = \sum_{n=1}^{\infty} m(E_n).$$

Proof Since $E_1 \in \mathcal{M}_d, \forall T \subset \mathbb{R}^d$,

$$m^*(T) = m^*(T \cap E_1) + m^*(T \cap E_1^c).$$

Since $E_2 \in \mathcal{M}_d$, we get that

$$\begin{aligned} m^*(T \cap E_1^c) &= m^*((T \cap E_1^c) \cap E_2) + m^*((T \cap E_1^c) \cap E_2^c) \\ &= m^*(T \cap (E_1^c \cap E_2)) + m^*(T \cap (E_1 \cup E_2)^c). \end{aligned}$$

Since $E_1 \cap E_2 = \emptyset$, $E_1^c \cap E_2 = E_2$. Thus,

$$m^*(T) = m^*(T \cap E_1) + m^*(T \cap E_2) + m^*(T \cap (E_1 \cup E_2)^c).$$

Then since $E_3 \in \mathcal{M}_d$,

$$\begin{aligned} m^*(T \cap (E_1 \cup E_2)^c) &= m^*(T \cap (E_1 \cup E_2)^c \cap E_3) + m^*(T \cap (E_1 \cup E_2)^c \cap E_3^c) \\ &= m^*(T \cap (E_1^c \cap E_2^c \cap E_3)) + m^*(T \cap (E_1 \cup E_2 \cup E_3)^c). \end{aligned}$$

Since $E_1 \cap E_3 = \emptyset$ and $E_2 \cap E_3 = \emptyset$, $E_1^c \cap E_2^c \cap E_3 = E_3$. Then

$$m^*(T) = m^*(T \cap E_1) + m^*(T \cap E_2) + m^*(T \cap E_3) + m^*(T \cap (E_1 \cup E_2 \cup E_3)^c).$$

Continuing this process, we get

$$m^*(T) = \sum_{n=1}^N m^*(T \cap E_n) + m^*\left(T \cap \left(\bigcup_{n=1}^N E_n\right)^c\right), \quad \forall N \in \mathbb{N}_+.$$

Since $\bigcup_{n=1}^N E_n \subseteq \bigcup_{n=1}^{\infty} E_n = E$, the monotonicity gives that

$$m^*(T) \geq \sum_{n=1}^N m^*(T \cap E_n) + m^*(T \cap E^c), \quad \forall N \in \mathbb{N}_+.$$

Letting $N \rightarrow \infty$, we get that

$$\begin{aligned} m^*(T) &\geq \sum_{n=1}^{\infty} m^*(T \cap E_n) + m^*(T \cap E^c) \\ &\geq m^*(T \cap E) + m^*(T \cap E^c). \end{aligned} \tag{11.4}$$

Then by subadditivity of outer measure,

$$m^*(T) = m^*(T \cap E) + m^*(T \cap E^c).$$

Hence E is measurable. Substituting $T = E$ into (11.4), we get

$$m(E) = \sum_{n=1}^{\infty} m(E_n).$$

□

Remark 1. The additivity of Lebesgue measure also holds for the union of finitely many measurable sets. Suppose $E_1, E_2, \dots, E_k$ are disjoint and set $E_j = \emptyset$ for $j > k$. Then $\{E_j : j \in \mathbb{N}_+\}$ are disjoint. So $\bigcup_{j=1}^k E_j = \bigcup_{j=1}^{\infty} E_j \in \mathcal{M}_d$ and

$$m\left(\bigcup_{j=1}^k E_j\right) = m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j) = \sum_{j=1}^k m(E_j).$$

2. Additivity of measure implies the monotonicity of measure. If E and F are both measurable and $F \subseteq E$, then $m(E) - m(F) = m(E \setminus F) \geq 0$.

Corollary 11.3.15 (Subadditivity) Let $\{E_n\}$ be a sequence of measurable sets contained in $\mathbb{R}^d$. Then $E = \bigcup_{n=1}^{\infty} E_n$ is measurable. And

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m(E_n).$$

Proof Since $E_1 \cup E_2$ is the disjoint union of E_1 and $E_2 \setminus E_1$, $E_1 \cup E_2 \in \mathcal{M}_d$. Then by induction, $\bigcup_{k=1}^{n-1} E_k \in \mathcal{M}_d$ for all $n \geq 2$.

Set $F_n = E_n \setminus \left(\bigcup_{k=1}^{n-1} E_k\right)$. Then $\{F_n : n \in \mathbb{N}_+\}$ are disjoint measurable sets. By Theorem 11.3.14, $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n \in \mathcal{M}_d$. And since $F_n \subseteq E_n$,

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = m\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} m(F_n) \leq \sum_{n=1}^{\infty} m(E_n).$$

□

Let $\mathcal{F}$ be a collection of sets. We say $\mathcal{F}$ is a σ -algebra if

1. $\emptyset, \mathbb{R}^d \in \mathcal{F}$;
2. $E, F \in \mathcal{F}$ implies $E \setminus F \in \mathcal{F}$;
3. $\bigcup_{n=1}^{\infty} E_n \in \mathcal{F}$ whenever $E_n \in \mathcal{F}$ for all $n \in \mathbb{N}_+$.

By the results of Example 11.3.11, Proposition 11.3.13 and Corollary 11.3.15, we know that $\mathcal{M}_d$ forms a σ -algebra.

Remark 1. $\mathcal{M}_d$ is also closed under finite union. Let $E_1, E_2, \dots, E_k \subseteq \mathbb{R}^d$. We set $E_j = \emptyset$ for $j > k$. Then $\bigcup_{j=1}^k E_j = \bigcup_{j=1}^{\infty} E_j \in \mathcal{M}_d$.

2. The collection $\mathcal{M}_d$ is closed under intersection. Let $\{E_n\}$ be a sequence of disjoint measurable sets. Then by De Morgan's Law,

$$\bigcap_{n=1}^N E_n = \left(\bigcup_{n=1}^N E_n^c\right)^c \in \mathcal{M}_d \quad \text{and} \quad \bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^c\right)^c \in \mathcal{M}_d.$$

Example 11.3.16 All open subsets and closed subsets of $\mathbb{R}^d$ are measurable.

Proof Let E be a nonempty open set. By Theorem 11.2.13, there exists countably infinitely many dyadic cubes Q_n such that $E = \bigcup_{n=1}^{\infty} Q_n$.

By the result of Example 11.3.12, $Q_n \in \mathcal{M}_d$ for all $n \in \mathbb{N}_+$. Then by Corollary 11.3.15, $E \in \mathcal{M}_d$.

If F is a closed set, then F^c is open. Since $F^c \in \mathcal{M}_d$, $F = (F^c)^c \in \mathcal{M}_d$. □

The next result tells us every measurable set is nearly an open set.

Proposition 11.3.17 Let $E \in \mathcal{M}_d$, then $\forall \varepsilon > 0$, there exists an open set $G \subseteq \mathbb{R}^d$ such that

$$E \subseteq G \quad \text{and} \quad m(G \setminus E) < \varepsilon.$$

Proof If $m(E) = m^*(E) < +\infty$, then $\forall \varepsilon > 0$, there exists a sequence of open cuboids $\{Q_n\}$ such that

$$E \subseteq \bigcup_{n=1}^{\infty} Q_n \quad \text{and} \quad \sum_{n=1}^{\infty} |Q_n| < m(E) + \varepsilon.$$

Let $G = \bigcup_{n=1}^{\infty} Q_n$, then G is open and

$$m(G) = m\left(\bigcup_{n=1}^{\infty} Q_n\right) \leq \sum_{n=1}^{\infty} |Q_n| < m(E) + \varepsilon.$$

This implies that $m(G \setminus E) < \varepsilon$.

If $m(E) = +\infty$, let $E_n = E \cap B_n(O)$ for all $n \in \mathbb{N}_+$. Then $m(E_n) < +\infty$. So there exists open set $G_n \subseteq \mathbb{R}^d$ such that $E_n \subseteq G_n$ and $m(G_n \setminus E_n) < 2^{-n}\varepsilon$.

Set $G = \bigcup_{n=1}^{\infty} G_n$, then G is open and $E \subseteq G$. Since

$$G \setminus E = \left(\bigcup_{n=1}^{\infty} G_n\right) \cap E^c = \bigcup_{n=1}^{\infty} (G_n \cap E^c) \subseteq \bigcup_{n=1}^{\infty} (G_n \cap E_n^c) = \bigcup_{n=1}^{\infty} (G_n \setminus E_n),$$

we get

$$m(G \setminus E) \leq \sum_{n=1}^{\infty} m(G_n \setminus E_n) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon.$$

□

Corollary 11.3.18 Let $E \in \mathcal{M}_d$, then $\forall \varepsilon > 0$, there exists a closed set $F \subseteq E$ such that

$$m(E \setminus F) < \varepsilon.$$

Proof $\forall \varepsilon > 0$, there exists an open set G such that $E^c \subseteq G$ and $m(G \setminus E^c) < \varepsilon$. Let $F = G^c$, then F is closed and $F \subseteq E$. Since $E \setminus F = E \cap F^c = E \cap G = G \setminus E^c$,

$$m(E \setminus F) = m(G \setminus E^c) < \varepsilon.$$

□

At the end of this section, we want to introduce one more property of Lebesgue measure.

Proposition 11.3.19 Let $\{E_n\}$ be a sequence of measurable sets of $\mathbb{R}^d$.

1. If $E_n \subset E_{n+1}$ for all $n \in \mathbb{N}_+$, then

$$\lim_{n \rightarrow \infty} m(E_n) = m\left(\bigcup_{n=1}^{\infty} E_n\right).$$

2. If $m(E_1) < +\infty$ and $E_{n+1} \subset E_n$ for all $n \in \mathbb{N}_+$, then

$$\lim_{n \rightarrow \infty} m(E_n) = m\left(\bigcap_{n=1}^{\infty} E_n\right).$$

Proof 1. Set $F_1 = E_1$, $F_{n+1} = E_{n+1} \setminus E_n$ for all $n \in \mathbb{N}_+$. Then $\{F_n : n \in \mathbb{N}_+\}$ are disjoint, and $E_n = \bigcup_{k=1}^n F_k$. Thus,

$$m\left(\bigcup_{n=1}^{\infty} F_n\right) = \sum_{n=1}^{\infty} m(F_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n m(F_k) = \lim_{n \rightarrow \infty} m\left(\bigcup_{k=1}^n F_k\right) = \lim_{n \rightarrow \infty} m(E_n).$$

2. Let $G_n = E_1 \setminus E_n$, then $G_n \subset G_{n+1}$ for all $n \in \mathbb{N}_+$. And

$$E_1 \setminus \bigcap_{n=1}^{\infty} E_n = E_1 \cap \left(\bigcap_{n=1}^{\infty} E_n\right)^c = E_1 \cap \left(\bigcup_{n=1}^{\infty} E_n^c\right) = \bigcup_{n=1}^{\infty} (E_1 \cap E_n^c) = \bigcup_{n=1}^{\infty} G_n.$$

Then

$$\begin{aligned} m(E_1) - m\left(\bigcap_{n=1}^{\infty} E_n\right) &= m\left(E_1 \setminus \bigcap_{n=1}^{\infty} E_n\right) = m\left(\bigcup_{n=1}^{\infty} G_n\right) = \lim_{n \rightarrow \infty} m(G_n) \\ &= \lim_{n \rightarrow \infty} [m(E_1) - m(E_n)] = m(E_1) - \lim_{n \rightarrow \infty} m(E_n). \end{aligned}$$

Hence $\lim_{n \rightarrow \infty} m(E_n) = m\left(\bigcap_{n=1}^{\infty} E_n\right)$. □

11.4 Mapping on $\mathbb{R}^d$

In this section, we are devoted to study mappings between two real spaces.

Definition 11.4.1 A mapping $\mathbf{f}$ (or vector-valued function) is a relation between a set $A \subseteq \mathbb{R}^p$ and a set $B \subseteq \mathbb{R}^q$ that associates to every element of A exactly one element of B . A is called the domain of $\mathbf{f}$, B is called the codomain of $\mathbf{f}$, and

$$\mathbf{f}(A) := \{\mathbf{f}(\vec{x}) : \vec{x} \in A\}$$

is called the range or the map of $\mathbf{f}$.

There are some basic properties of mappings.

Exercises

1. Let $S \subseteq \mathbb{R}^d$. What are the definitions of inner points of S , limit points of S , $\mathring{S}$, S^c , ∂S , $\overline{S}$ and S' ? What are the relationships between them?

2. What are the definitions and structures of open and closed sets?

3. How to describe the completeness of $\mathbb{R}^d$?

4. Summarize the properties of outer measure. Which properties coincide with the properties of length, area and volume? Which ones are not?

5. Find the derive sets and boundary sets of the following sets.

(1) $[a, b) \times [c, d]$;

(2) $\{(x, y) : xy \neq 0\}$;

(3) $\mathbb{Z} \times \mathbb{Z}$;

(4) $\{(x, y) : y = \sin \frac{1}{x}\}$.

6. Show that if F_1 and F_2 are both closed, then $F_1 \cup F_2$ is closed.

7. Show that if $\{F_\alpha : \alpha \in I\}$ are all closed sets, then $\bigcap_{\alpha \in I} F_\alpha$ is closed.

8. Let $S \subseteq \mathbb{R}^d$. Show that $(\mathring{S})^c = \overline{S^c}$.

9. Let $A \subseteq \mathbb{R}^d$ with $m^*(A) = 0$. Show that $\forall B \subseteq \mathbb{R}^d$,

$$m^*(A \cup B) = m^*(B \setminus A) = m^*(B).$$

10. Let $E \subseteq \mathbb{R}^1$ and $\lambda \in \mathbb{R}$. Set $\lambda E := \{\lambda x : x \in E\}$. Show that

$$m^*(\lambda E) = |\lambda| \cdot m^*(E).$$

11. For $E \subseteq \mathbb{R}^d$ and $\delta > 0$, set

$$m_\delta^*(E) = \inf \left\{ \sum_{n=1}^{\infty} |Q_n| : Q_n \text{ are open cuboids with side-length less than } \delta, E \subseteq \bigcup_{n=1}^{\infty} Q_n \right\}.$$

Show that $m_\delta^*(E) = m^*(E)$.

12. Let $E \subseteq [a, b]$ with $m^*(E) > 0$. Show that $\forall c \in (0, m^*(E))$, $\exists A \subseteq E$ such that $m^*(A) = c$.

13. Let $E \subseteq \mathbb{R}^d$ with $m(E) = 0$. Show that $\mathring{E} = \emptyset$.

14. Let $\mathbf{f}$ be a mapping from $\mathbb{R}^p$ to $\mathbb{R}^q$. And for each open set $A \subseteq \mathbb{R}^q$, $\mathbf{f}^{-1}(A)$ is open in $\mathbb{R}^p$. Show that $\mathbf{f}$ is continuous.

15. Let $D = \{(x, y, 0) : (x, y) \in \mathbb{R}^2\}$ and $E = \{(x, y, z) \in \mathbb{R}^3 : x = y\}$. Show that $D, E \in \mathcal{M}_3$ and $m(D) = m(E) = 0$.

16. Let $A_1 \subseteq A_2 \subseteq \mathbb{R}^d$. Show that if $A_1 \in \mathcal{M}_d$ and $m^*(A_2) = m(A_1) < +\infty$, then $A_2 \in \mathcal{M}_d$.

17. Let $\{A_n : n \in \mathbb{N}_+\}$ be disjoint measurable sets in $\mathbb{R}^d$. Show that if $B_n \subseteq A_n$,

$$m^*\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} m^*(B_n).$$

18. Let $E \subseteq \mathbb{R}^d$. Show that

$$m^*(E) = \inf\{m(G) : G \text{ is an open set containing } E\}.$$

19. Let $E \subseteq \mathbb{R}^d$ with $m^*(E) < +\infty$. The inner measure of E is defined by

$$m_*(E) = \sup\{m(F) : F \text{ is a closed set contained in } E\}.$$

Show that if $m_*(E) = m^*(E)$, then $E \in \mathcal{M}_d$

20. Let $A \in \mathcal{M}_d$. Show that $\forall B \subseteq \mathbb{R}^d$,

$$m^*(A \cup B) + m^*(A \cap B) = m^*(A) + m^*(B).$$

21. Show that $\forall A, B \subseteq \mathbb{R}^d$,

$$m^*(A \cup B) + m^*(A \cap B) \leq m^*(A) + m^*(B).$$

22. Let $E \subseteq \mathbb{R}^d$. $\forall \varepsilon > 0$, there exists an open set $G \subseteq \mathbb{R}^d$ such that $E \subseteq G$ and $m^*(G \setminus E) < \varepsilon$. Show that $E \in \mathcal{M}_d$.

23. Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be three distinct points in $\mathbb{R}^2$. And let S be the set of points inside $\triangle ABC$. Show that $m(S) = 2^{-1}|\det T|$, where

$$T = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}.$$

Is $m(S)$ equal to the area of $\triangle ABC$?

24. We say a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous, if $\forall \varepsilon > 0, \exists \delta > 0$, for each collection of finitely many disjoint open intervals $\{(a_n, b_n) : n = 1, 2, \dots, N\}$,

$$\sum_{n=1}^N |b_n - a_n| < \delta \quad \text{implies} \quad \sum_{n=1}^N |f(b_n) - f(a_n)| < \varepsilon.$$

Show that

(1) if f satisfies Lipschitz condition, then f is absolutely continuous;

(2) if f is absolutely continuous, then f maps a set of measure zero to a set of measure zero.